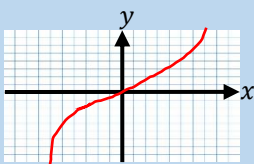


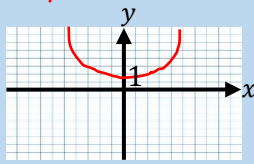
Hyperbolic trig functions

Key Points

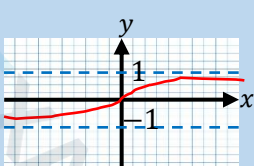
$$\sinh x = \frac{e^x - e^{-x}}{2}$$



$$\cosh x = \frac{e^x + e^{-x}}{2}$$



$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



Basic Skills

Q1. Find the value of

- a) $\sinh 2$ b) $\tanh \pi$ c) $\cosh -3$

Q2. Sketch the graphs stating the domain and range

- a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

Q3. Are the following statements true or false

- a) "Hyperbolic sine has a period of 2π "
b) "The maximum and minimum of hyperbolic tan is $-1/1$ "

Competency Skills

Q4. Standard trigonometry is aimed at satisfying the unit circle, hyperbolic trigonometry is aimed at satisfying the unit hyperbola

- a) Explain why $\cos^2 \theta + \sin^2 \theta = 1$ describes the unit circle
b) Write the equation of a hyperbola with asymptotes $y = \pm x$
c) Show that $\cosh^2 \theta - \sinh^2 \theta = 1$
d) Hence explain why $\cosh^2 \theta - \sinh^2 \theta = 1$ describes the unit hyperbola

Q5. Solve the following equations using exponential form definitions

- a) $\sinh x = 2$ b) $\cosh x = 6$ c) $\tanh x = \frac{3}{4}$

Q6. Prove that $\sinh 2x = 2 \cosh x \sinh x$

Advanced Skills

Q7. Sketch the following graphs stating the domain and range

- a) $y = 1 - \cosh^2 x$ b) $y = 2 \sinh 3x - 5$ c) $y = \tanh(3 - x^2)$

Q8. Evaluate $\lim_{x \rightarrow \infty} \tanh(-x)$

Q9. It is suggested that by looking at the graphs that we have $\sinh x \approx x^3$, $\cosh x \approx x^2 + 1$, determine if these estimations seem reasonable by calculating values.

Q10. By solving $\sinh y = x$ write down $y = \operatorname{arcsinh} x$ in logarithm form

Extension

Q11. Predict how the hyperbolic trig identities differ from normal trig identities

Hint: Consider its on the hyperbola not circle, what changes